

Lecture 19

Formula for determinant $\rightarrow n!$ terms.

$$\Rightarrow \begin{vmatrix} a & b \\ c & d \end{vmatrix} = \begin{vmatrix} a & 0 \\ c & d \end{vmatrix} + \begin{vmatrix} 0 & b \\ c & d \end{vmatrix}$$

$$\begin{vmatrix} a & 0 \\ 0 & d \end{vmatrix} + \cancel{\begin{vmatrix} a & 0 \\ c & 0 \end{vmatrix}} + \begin{vmatrix} 0 & b \\ c & 0 \end{vmatrix} + \cancel{\begin{vmatrix} 0 & b \\ 0 & d \end{vmatrix}}$$

$$\Rightarrow \begin{vmatrix} a & 0 \\ 0 & d \end{vmatrix} + \begin{vmatrix} 0 & b \\ c & 0 \end{vmatrix} = \underline{\underline{ad - bc}}$$

The idea is the same for any $n \times n$ matrix.

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} \rightarrow \text{Split into 3 after first row.}$$

↳ Split into 9 after 2nd row

↳ Split into 27 after 3rd row

⇒ The only survivors after these three steps would be the one where one column is not all zero. And we know that after splitting we only have one element in each row

⇒ 6 total matrices left. for 3×3 !!

$$\Rightarrow \begin{vmatrix} a_{11} & & \\ & a_{22} & \\ & & a_{33} \end{vmatrix} + \begin{vmatrix} a_{11} & & \\ & & a_{23} \\ & & a_{32} \end{vmatrix}$$

$$+ \begin{vmatrix} 0 & a_{12} & 0 \\ a_{21} & 0 & 0 \\ 0 & 0 & a_{33} \end{vmatrix} + \begin{vmatrix} 0 & 0 & a_{13} \\ a_{21} & 0 & 0 \\ 0 & a_{32} & 0 \end{vmatrix}$$

$$+ \begin{vmatrix} 0 & & a_{13} \\ 0 & a_{22} & \\ a_{31} & & \end{vmatrix} + \begin{vmatrix} & a_{12} & \\ & & a_{23} \\ a_{31} & & \end{vmatrix}$$

$$\Rightarrow a_{11} a_{22} a_{33} - a_{11} a_{23} a_{32} +$$

$$a_{13} a_{21} a_{32} - a_{12} a_{21} a_{33} +$$

$$a_{12} a_{23} a_{31} - a_{13} a_{22} a_{31}$$

$$\hookrightarrow \begin{matrix} a_{11} & (a_{22} a_{33} - a_{23} a_{32}) & + \\ a_{21} & (a_{13} a_{32} - a_{12} a_{33}) & + \\ a_{31} & (a_{12} a_{23} - a_{13} a_{22}) & \end{matrix}$$

→ determinant
of (n-1)
matrix!!

General formula $\rightarrow$

$$\det A = \sum_{n! \text{ terms}} \pm a_{1\alpha} a_{2\beta} a_{3\gamma} \dots a_{n\omega}$$

$(\alpha, \beta, \gamma \dots \omega) \rightarrow$ permutation of $(1 \dots n)$

$$\cong \begin{vmatrix} 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{vmatrix}$$

$$\Rightarrow \begin{vmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{vmatrix} \Rightarrow 0$$

⇒ Cofactors formula

This formula represents the $n \times n$ determinant as $(n-1)$ determinants.

We fix one row/column and operate on the remaining rows/cols one by one.

$$\det A_{3 \times 3} = a_{11} (a_{22} a_{33} - a_{23} a_{32}) + a_{12} (a_{23} a_{31} - a_{21} a_{33}) + a_{13} (a_{21} a_{32} - a_{22} a_{31})$$

$$\begin{aligned} \Rightarrow \text{Cofactor of } a_{ij} &\Rightarrow \pm \det \left(\begin{array}{c} n-1 \text{ matrix} \\ \text{without } i \text{ row} \\ j \text{ column} \end{array} \right) \\ &= C_{ij} \\ &= (-1)^{i+j} \det \left(\begin{array}{c} \text{matrix} \\ \text{without } i \text{ row} \\ j \text{ column} \end{array} \right) \end{aligned}$$

$$\det A = a_{11} C_{11} + a_{12} C_{12} \dots a_{1n} C_{1n}$$

along row 1

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

$$\begin{aligned} \det |A_{2 \times 2}| &= (-1)^{1+1} a |d| + (-1)^{1+2} b |c| \\ &= \underline{\underline{ad - bc}} \end{aligned}$$

eg

$$A_n = \begin{vmatrix} i_1 & 1 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{vmatrix}$$

$$|A_1| = 1$$

$$|A_2| = 0$$

$$|A_3| = -1$$

$$1 \cdot (1 \cdot 0 - 1) + (-1) \cdot 0 = \underline{\underline{-1}} = |A_n|$$

$$|A_n| = |A_{n-1}| - |A_{n-2}| \text{ for tridiagonal matrices! }$$